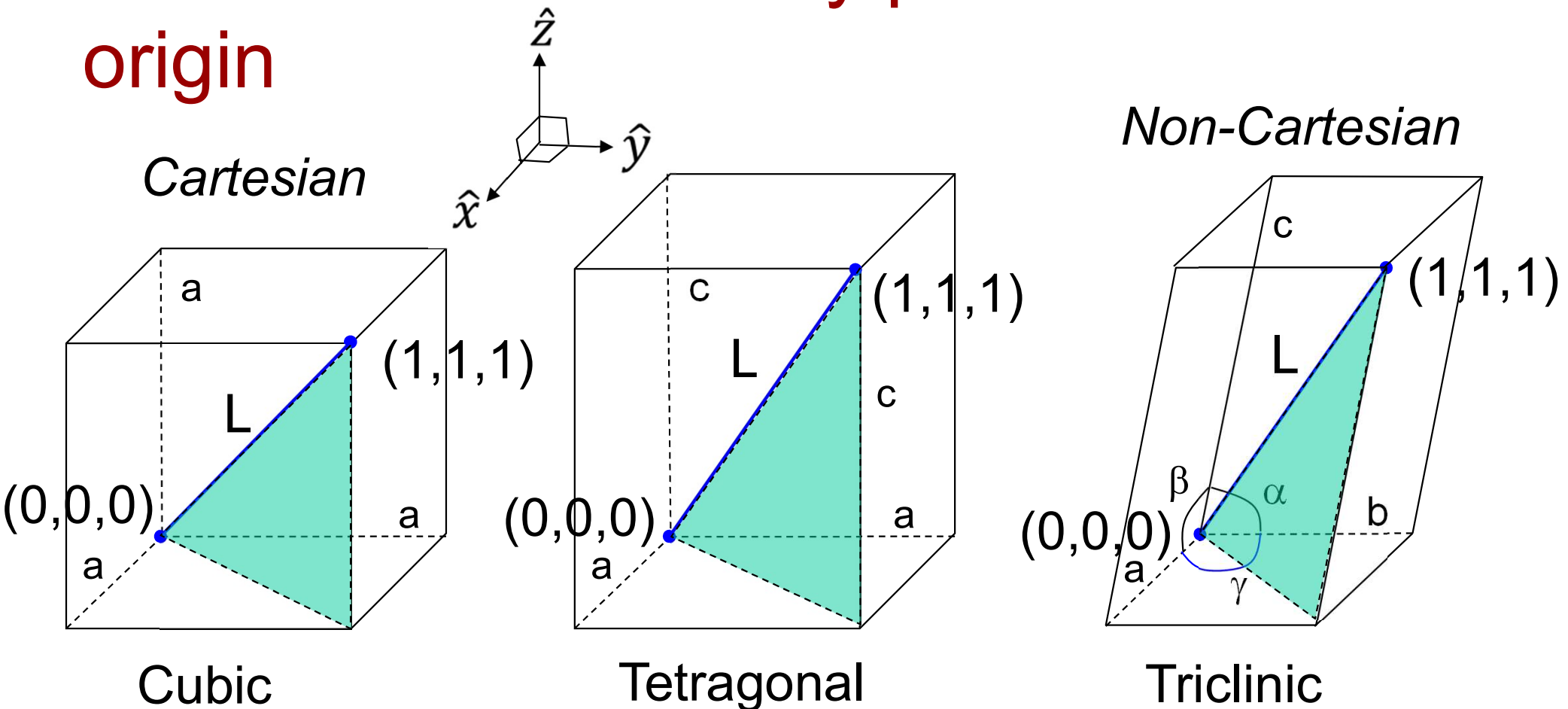


Topic 7: Crystallographic Computations in 3D

Questions we will answer...

- What is the distance between any two points in an arbitrary Bravais lattice?
- What is the angle between two vectors?
- What is a metric tensor and why is it useful to calculating the above?
- What is the volume of a 3D Bravais unit cell?

Distance between any point and the origin



$$L = \sqrt{(1 \cdot a)^2 + (1 \cdot a)^2 + (1 \cdot a)^2}$$

$$= a\sqrt{3}$$

$$L = \sqrt{(1 \cdot a)^2 + (1 \cdot a)^2 + (1 \cdot c)^2}$$

$$= \sqrt{2a^2 + c^2}$$

$L = ?$

Pythagoras' theorem works in Cartesian (orthonormal) reference frames

In-Class Exercise 1

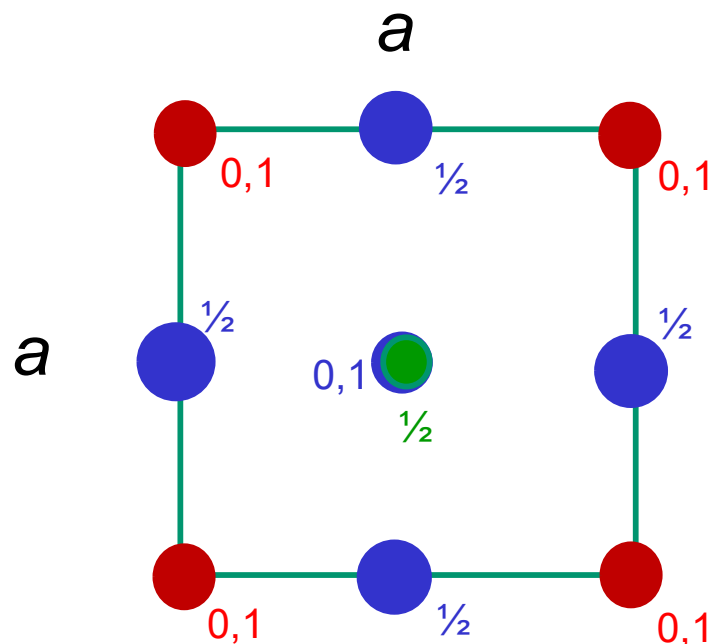
The unit cell of an oxide crystal with a cubic lattice parameter of 4.00 Å contains the following atoms (the ordered sets are the coefficients of location that define r_j):

A at (0,0,0) ; B at (1/2 , 1/2, 1/2);

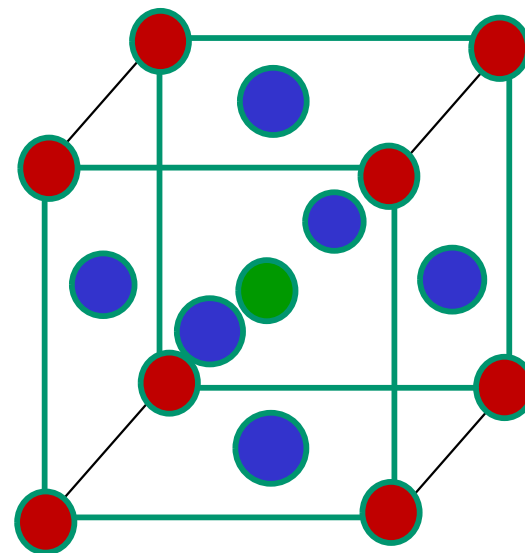
O at (1/2, 1/2, 0); O at (0, 1/2, 1/2); O at (1/2, 0, 1/2).

- (i) Sketch projection and perspective views of the unit cell, including all atoms that are within or partially within the cell
- (ii) What is the formula unit of this compound?
- (iii) Specify the lattice and the basis of this crystal
- (iv) The compound is a cubic perovskite belonging to the $m\bar{3}m$ point group. What is the space group of this perovskite?
- (v) What are the coordination numbers of A and B?
- (vi) Determine the A-O and B-O bond lengths
- (vii) If A and B are not the same size, which is smaller?
- (viii) Specify the sequence of atoms found along the $\langle 111 \rangle$ and $\langle 110 \rangle$ directions
- (ix) Specify the contents of the (200) plane and the contents of the (110) plane

In-Class Exercise 1 (solution)



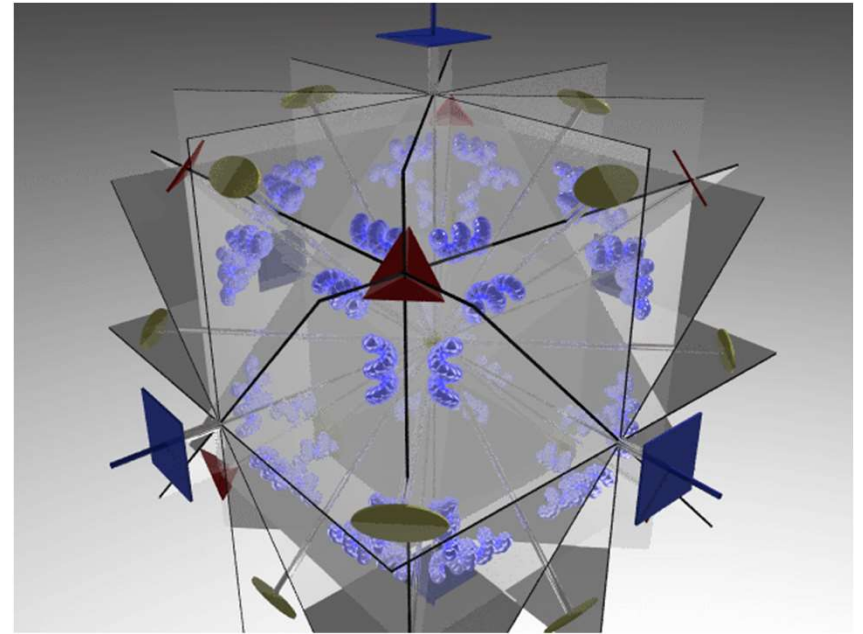
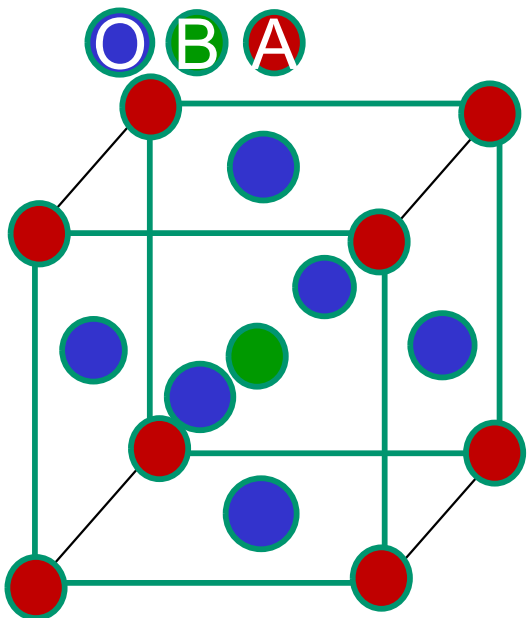
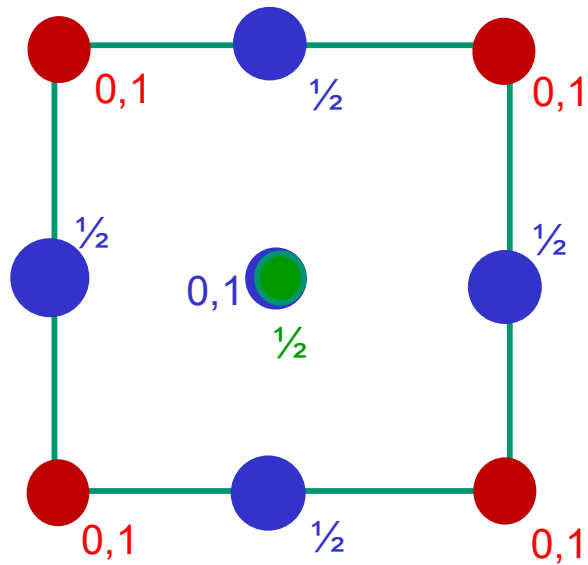
A at (0,0,0) ; B at
(1/2 , 1/2, 1/2);
O at (1/2, 1/2, 0);
O at (0, 1/2, 1/2);
O at (1/2, 0, 1/2).



A: $8 \times 1/8 = 1$ (corner)
B: 1 (center)
O: $6 \times 1/2 = 3$ (face)

- Formula unit is **ABO₃** (perovskite)
- One formula unit per unit cell
- **Cubic P** with **ABO₃ basis** at
A: (0,0,0), B: (1/2,1/2,1/2), O:
(1/2,1/2,0) (1/2,0,1/2) (0,1/2,1/2)
- Space group: **Pm $\bar{3}$ m**

In-Class Exercise 1 (solution)



https://www.xtal.iqfr.csic.es/Cristalografia/parte_03-en.html

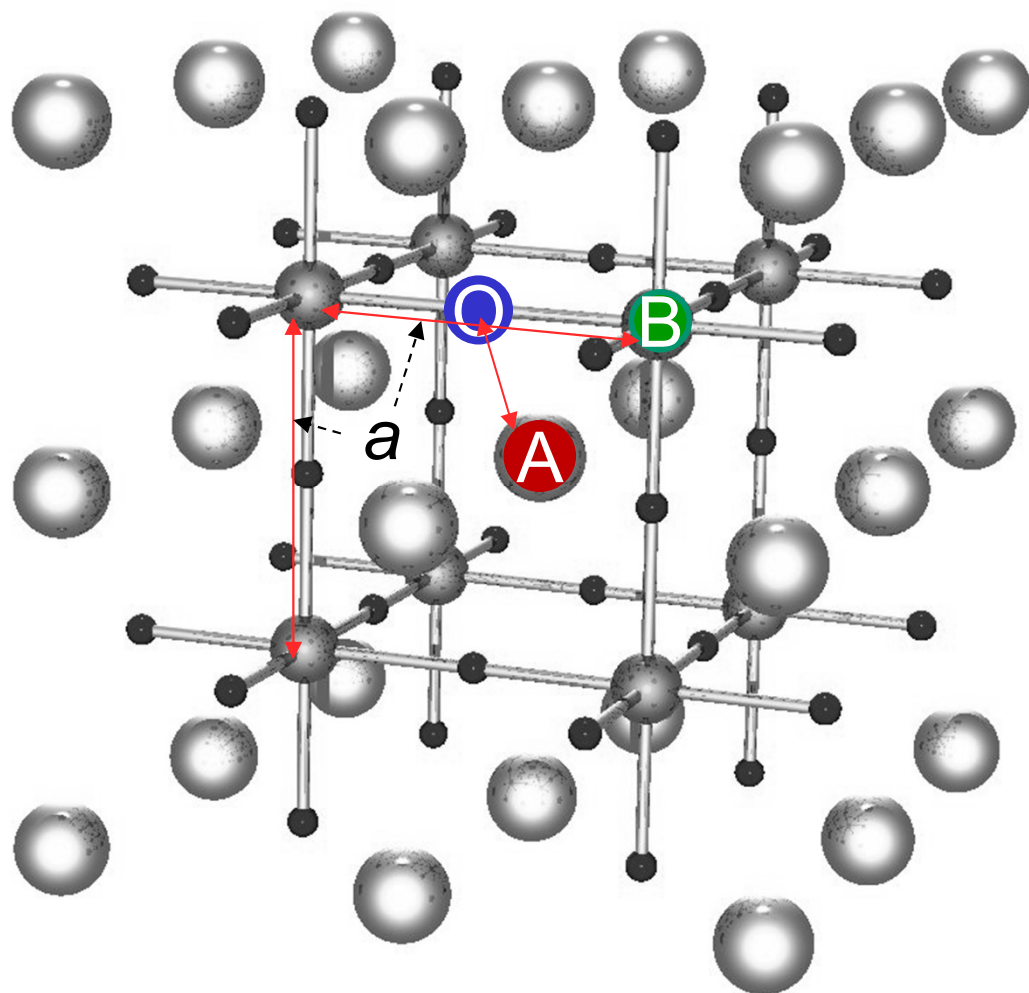
Crystal System	Point Groups	Space Groups		
Cubic	23	P23, P2 ₁ 3	F23	I23, I2 ₁ 3
	m $\bar{3}$	Pm $\bar{3}$, Pn $\bar{3}$, Pa $\bar{3}$	Fm $\bar{3}$, Fd $\bar{3}$	I $\bar{3}$, Ia $\bar{3}$
	432	P432, P4 ₂ 32, P4 ₃ 32, P4 ₁ 32	F432, F4 ₁ 32	I432, I4 ₁ 32
	43m	P43m, P43n	F43m, F43c	I43m, I43d
	m $\bar{3}m$	Pm $\bar{3}m$, Pn $\bar{3}n$, Pm $\bar{3}n$, Pn $\bar{3}m$	Fm $\bar{3}m$, Fm $\bar{3}c$, Fd $\bar{3}m$, Fd $\bar{3}c$	Im $\bar{3}m$, Ia $\bar{3}d$
Bravais lattice				



In-Class Exercise 1 (solution)

(Crystal systems with
orthogonal axes)

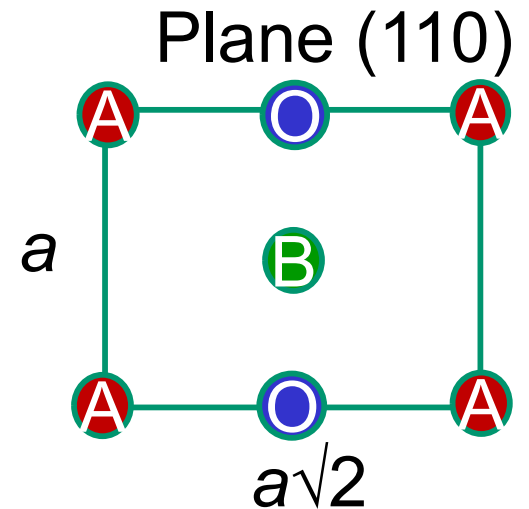
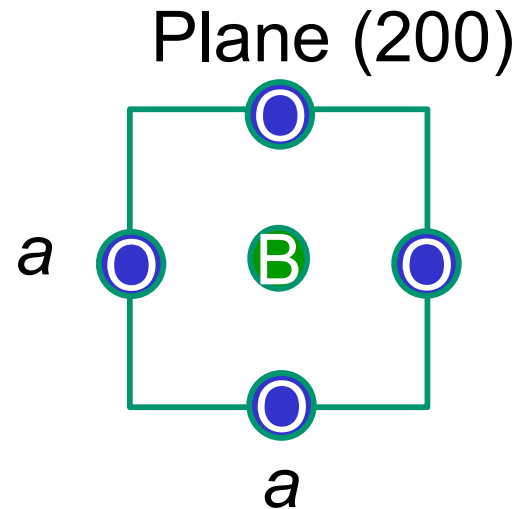
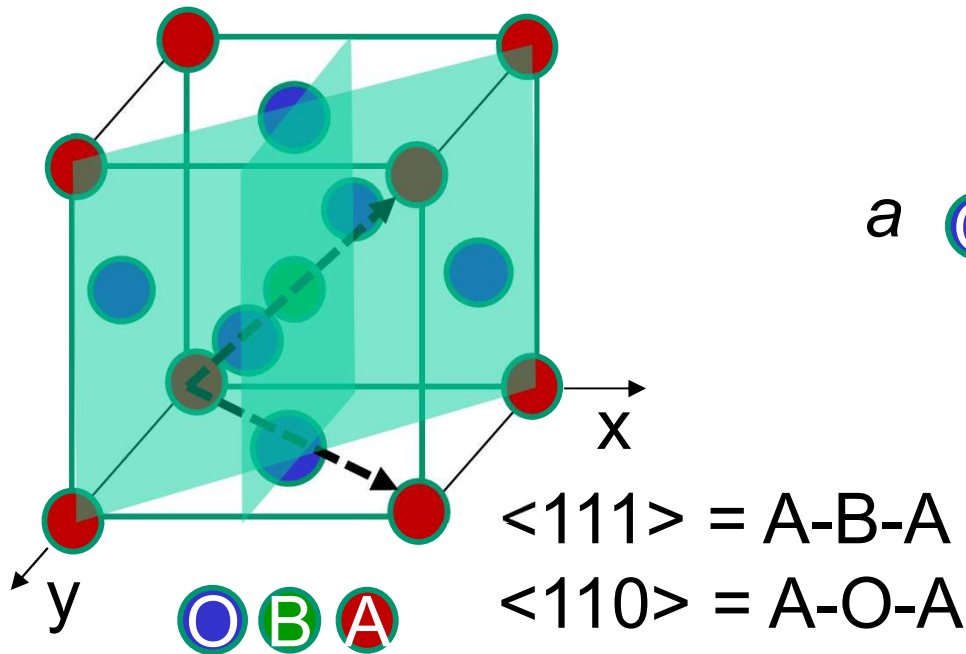
$$L^2 = (\Delta xa)^2 + (\Delta yb)^2 + (\Delta zc)^2$$



- The **B-O** bond length is $\frac{a}{2} = 2 \text{ \AA}$
- The **A-O** bond length is
$$L = a \sqrt{\left(\frac{1}{2}\right)^2 + 0 + \left(\frac{1}{2}\right)^2} = 4\text{\AA} \sqrt{\frac{1}{2}} = 2.8\text{\AA}$$
- Given that $d_{A-O} > d_{B-O}$, the A atom must be larger than the B atom

B atoms have CN = 6 (octahedron)
A atoms have CN = 12 (cuboctahedron)

In-Class Exercise 1 (solution)



Q: What is the angle between $\langle 111 \rangle$ and $\langle 110 \rangle$?

Q: What is the angle between (200) and (110)?

Q: What changes if crystal system is orthorhombic?

Q: What changes if crystal system is NOT orthogonal?

“A **perovskite** is any material with the same type of crystal structure as **calcium titanate** (CaTiO_3), known as the perovskite structure, or $\text{A}^{2+}\text{B}^{4+}(\text{X}^{2-})_3$ with the oxygen anion in the face centers.

Perovskites take their name from the mineral, which was first discovered in the Ural mountains of Russia by Gustav Rose in 1839 and is named after Russian mineralogist L. A. Perovski (1792–1856).“
 , in Wikipedia (retrieved 13 Sept 2017)

(Length of a vector)

$$|\vec{s}| = \sqrt{\vec{s} \cdot \vec{s}}$$

Vector Operations

(angle between two vectors)

$$\vec{s} \cdot \vec{t} = |\vec{s}| |\vec{t}| \cos \theta$$

$$\cos \theta = \frac{\vec{s} \cdot \vec{t}}{|\vec{s}| |\vec{t}|}$$

s_i and t_i are expressed in terms of **Miller indices**:

- distance ($\Delta x, \Delta y, \Delta z$)
- Direction $[h \ k \ l]$

$$\vec{s} \cdot \vec{t} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} [\Gamma] \begin{bmatrix} t_1 & t_2 & t_3 \end{bmatrix}$$

$[\Gamma]$: Metric tensor, contains ALL **unit cell** info and **units**

We need **unit cell** info ($a, b, c, \alpha, \beta, \gamma$) to compute actual distances

$$|\vec{s}| = \sqrt{\vec{s} \cdot \vec{s}} = \sqrt{\begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} [\Gamma] \begin{bmatrix} s_1 & s_2 & s_3 \end{bmatrix}}$$

Non-orthogonal:

$[\Gamma]$: non-diagonal matrix

Orthogonal:

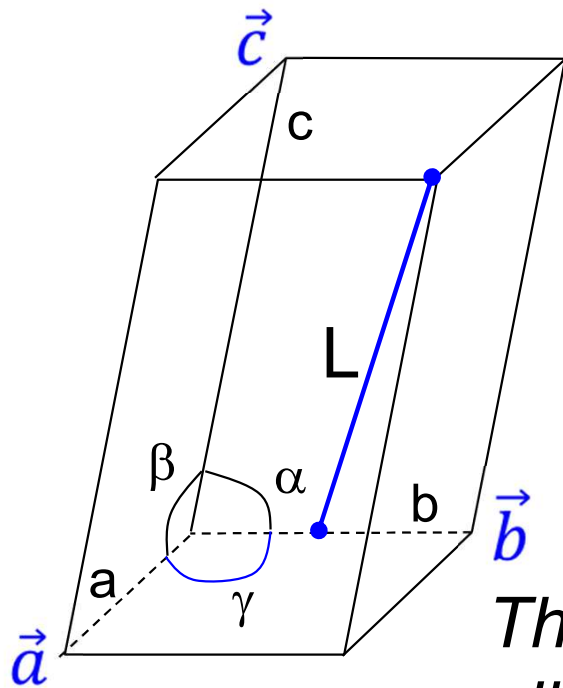
$[\Gamma]$: diagonal matrix

Cubic:

$[\Gamma]$: identity matrix $\times a^2$

$$\Gamma = \begin{bmatrix} a^2 & ab \cos \gamma & ac \cos \beta \\ ab \cos \gamma & b^2 & bc \cos \alpha \\ ac \cos \beta & bc \cos \alpha & c^2 \end{bmatrix}$$

Metric tensor in 3D



$$\Gamma \equiv \begin{bmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{bmatrix}$$

$$\Gamma = \begin{bmatrix} a^2 & ab \cos \gamma & ac \cos \beta \\ ab \cos \gamma & b^2 & bc \cos \alpha \\ ac \cos \beta & bc \cos \alpha & c^2 \end{bmatrix} \begin{matrix} \text{Most general} \\ \text{case:} \\ \text{triclinic} \end{matrix}$$

The metric tensor contains all the lattice parameters

Length, L , of a vector $\vec{R}$ with components $x\vec{a}$, $y\vec{b}$, $z\vec{c}$ is obtained by the scalar product:

$$L = (\vec{R} \cdot \vec{R})^{1/2}$$

Define vector $\vec{R}$ and metric tensor

$$L^2 = [x \ y \ z] \begin{bmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = [x \ y \ z][\Gamma] \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Result is a scalar

The metric tensor allows a **wide range of calculations*** to be performed correctly in orthogonal and non-orthogonal crystal systems

*We can compute:

- Distance between atoms and bond lengths (nearest neighbors)
- Angle of a bond and between two directions
- Volume of a unit cell

Metric tensor for all Bravais lattices

$$\Gamma_{\text{cubic}} = \begin{bmatrix} a^2 & 0 & 0 \\ 0 & a^2 & 0 \\ 0 & 0 & a^2 \end{bmatrix}$$

Identity matrix $\times a^2$

$$\Gamma_{\text{tetragonal}} = \begin{bmatrix} a^2 & 0 & 0 \\ 0 & a^2 & 0 \\ 0 & 0 & c^2 \end{bmatrix}$$

$$\Gamma_{\text{orthorhombic}} = \begin{bmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & c^2 \end{bmatrix}$$

$$L^2 = (\Delta x a)^2 + (\Delta y b)^2 + (\Delta z c)^2$$

(Crystal systems with
orthogonal axes)

$$\Gamma_{\text{hexagonal}} = \begin{bmatrix} a^2 & -\frac{a^2}{2} & 0 \\ -\frac{a^2}{2} & a^2 & 0 \\ 0 & 0 & c^2 \end{bmatrix}$$

$$\Gamma_{\text{monoclinic}} = \begin{bmatrix} a^2 & 0 & ac \cos \beta \\ 0 & b^2 & 0 \\ ca \cos \beta & 0 & c^2 \end{bmatrix}$$

$$\Gamma_{\text{rhombohedral}} = \begin{bmatrix} a^2 & a^2 \cos \alpha & a^2 \cos \alpha \\ a^2 \cos \alpha & a^2 & a^2 \cos \alpha \\ a^2 \cos \alpha & a^2 \cos \alpha & a^2 \end{bmatrix}$$

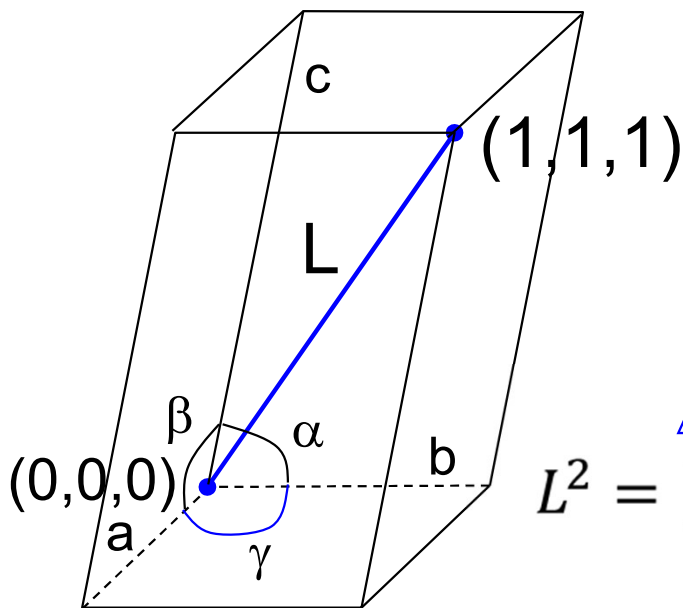
$$\Gamma_{\text{triclinic}} = \begin{bmatrix} a^2 & ab \cos \gamma & ac \cos \beta \\ ab \cos \gamma & b^2 & bc \cos \alpha \\ ac \cos \beta & cb \cos \alpha & c^2 \end{bmatrix}$$

$$L^2 = [(\Delta x a)^2 + (\Delta y b)^2 + (\Delta z c)^2 + 2ab\Delta x \Delta y \cos \gamma + 2ac\Delta x \Delta z \cos \beta + 2bc\Delta y \Delta z \cos \alpha].$$

(Crystal systems with
non-orthogonal axes)

Metric tensor in 3D

Distance between two points at (x_1, y_1, z_1) and (x_2, y_2, z_2) is the length of a vector $\vec{R}$ with components $(x_2 - x_1)\mathbf{a}$, $(y_2 - y_1)\mathbf{b}$ and $(z_2 - z_1)\mathbf{c}$.



$$L^2 = [\Delta x \quad \Delta y \quad \Delta z][\Gamma] \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

Length of Triclinic unit cell along [111]:

$$L^2 = \begin{bmatrix} \Delta x & \Delta y & \Delta z \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a^2 & ab \cos \gamma & ac \cos \beta \\ ab \cos \gamma & b^2 & bc \cos \alpha \\ ac \cos \beta & cb \cos \alpha & c^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

Triclinic

$$L^2 = a^2 + b^2 + c^2 + 2ab \cos \gamma + 2ac \cos \beta + 2bc \cos \alpha$$

$$L = \sqrt{a^2 + b^2 + c^2 + 2ab \cos \gamma + 2ac \cos \beta + 2bc \cos \alpha}$$

In-Class Exercise 2

Compute the distance between two atoms

Given: Crystal with lattice parameters $\{2, 2, 3, 90, 90, 90\}$ contains atoms at points $(1/2, 1/3, 1/4)$ and $(1/3, 1/2, 3/4)$.

Compute the distance between these two atoms

In-Class Exercise 3

Compute the angle between atomic bonds

Given: In a cubic crystal with lattice parameter a , an oxygen atom is present at the position $(0, 0, 0)$; this atom is bonded with two titanium atoms, located at the positions $(1/2, 1/2, 0)$, and $(1/2, 0, 1/2)$.

Compute the angle between these two bonds

Metric tensor and dot product

$$\vec{P} \cdot \vec{Q} = \sum_{i,j=1}^3 (P_i \vec{a}_i) \cdot (Q_j \vec{a}_j) = \sum_{i,j=1}^3 P_i \Gamma_{ij} Q_j = P_i \Gamma_{ij} Q_j$$

$$\{1, 2, 3\} = \{x, y, z\}$$

In a Cartesian coordinate: $\Gamma_{ij} = \delta_{ij}$, i.e., the identity matrix

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{P} \cdot \vec{Q} = P_i \Gamma_{ij} Q_j = P_i \delta_{ij} Q_j = P_i Q_i = P_1 Q_1 + P_2 Q_2 + P_3 Q_3$$

Summation implied (mixed products = 0) $P_x Q_x + P_y Q_y + P_z Q_z$

$$\vec{P} \cdot \vec{Q} = P_i Q_i$$

$$\vec{P} \cdot \vec{P} = P_i P_i$$

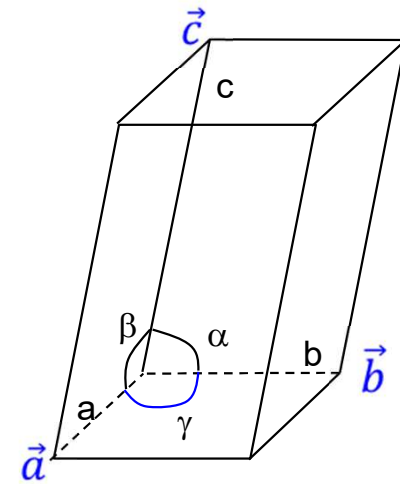
$$|\vec{P}| = \sqrt{P_i P_i}$$

In-Class Exercise 4

Compute the length of a vector

Given: Crystal with lattice parameters $\{3, 4, 6, 90, 120, 90\}$.

Compute the length of the main body diagonal



$\{a, b, c, \alpha, \beta, \gamma\}$

In-Class Exercise 5

Computation of the angle between lattice directions

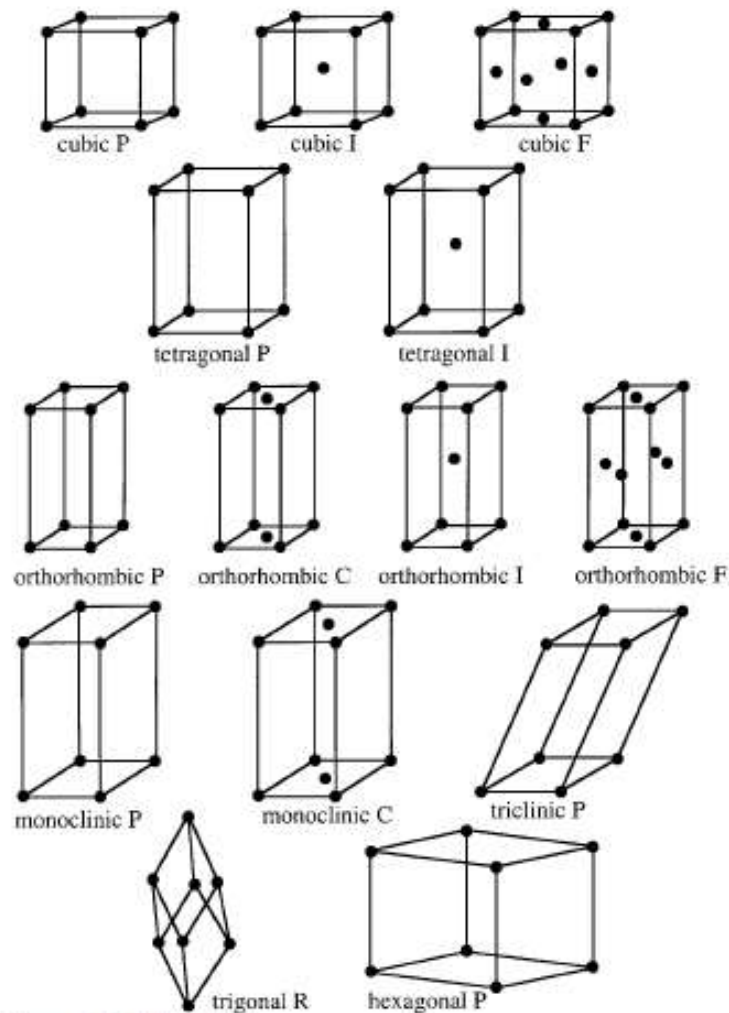
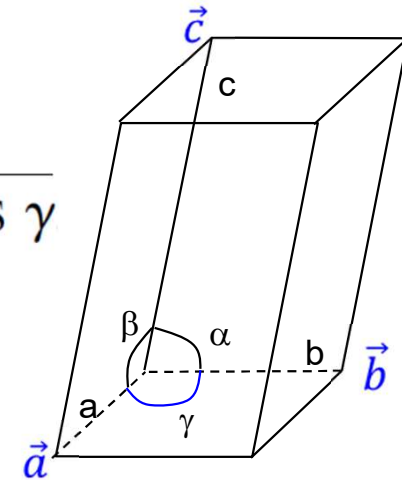
Given: Consider the monoclinic crystal with lattice parameters $a = 4$ nm, $b = 6$ nm, $c = 5$ nm and $\beta = 120^\circ$.

Compute the angle between the $[101]$ and $[\bar{2}01]$ directions

Volume of a Unit Cell

$$V = \vec{a} \cdot (\vec{b} \times \vec{c}) = abc \sqrt{1 - \cos^2 \alpha - \cos^2 \beta - \cos^2 \gamma + 2 \cos \alpha \cos \beta \cos \gamma}$$

Mixed vector product



cubic

$$V = a^3$$

tetragonal

$$V = a^2 c$$

orthorhombic

$$V = abc$$

monoclinic

$$V = abc \sin \beta$$

hexagonal

$$V = \frac{\sqrt{3}}{2} a^2 c$$

rhombohedral

$$V = a^3 \sqrt{1 - 3 \cos^2 \alpha + 2 \cos^3 \alpha}$$

triclinic $V = abc \sqrt{1 - \cos^2 \alpha - \cos^2 \beta - \cos^2 \gamma + 2 \cos \alpha \cos \beta \cos \gamma}$

Figure 2.12. The 14 Bravais lattices.

Volume of a Unit Cell

$$V = \vec{a} \cdot (\vec{b} \times \vec{c})$$

Determinant of
a 3x3 matrix

Alternative approach:

$$V = |\Gamma|^{\frac{1}{2}}$$

Square root of determinant
of metric tensor

$$V = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} \longrightarrow \vec{a} = a_x \hat{x} + a_y \hat{y} + a_z \hat{z}$$

Vertical lines
symbolize
determinant
operation

Tetragonal example:

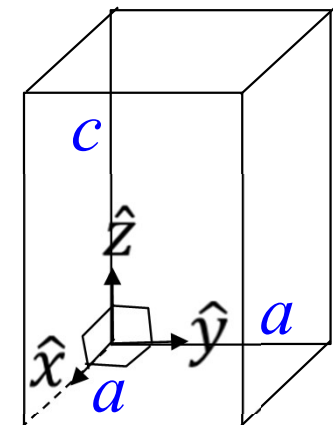
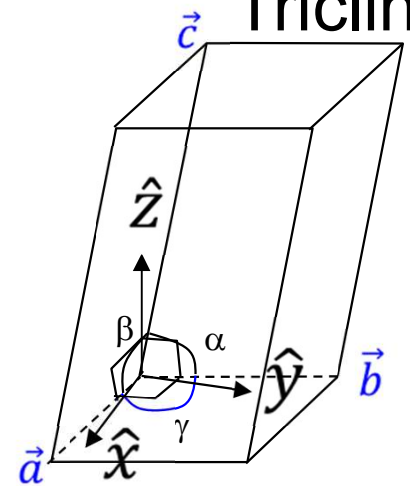
$$a=b \neq c; \alpha=\beta=\gamma=90^\circ$$

$$\vec{a} = a\hat{x} \quad \vec{b} = a\hat{y} \quad \vec{c} = c\hat{z}$$

$$V = \begin{vmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & c \end{vmatrix} = a^2 c$$

Non-Cartesian

Triclinic

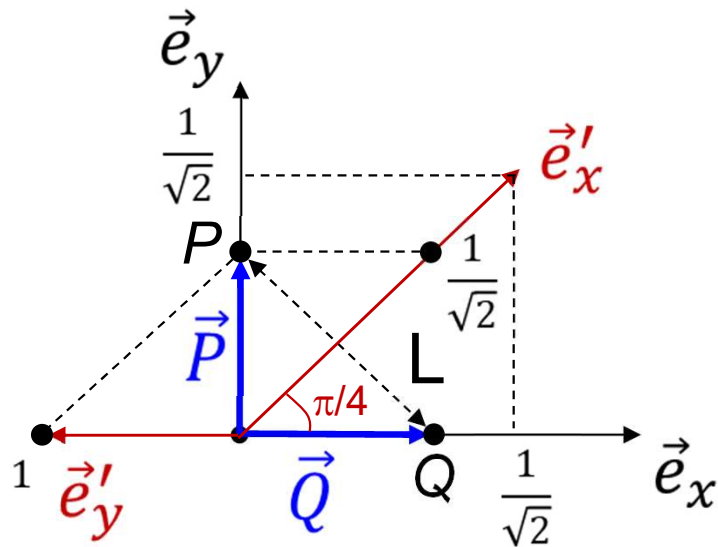


Additional considerations

This is bonus material showing the necessity for the need for a metric tensor approach when computing distances in Cartesian and non-cartesian references frames:

- Conversion between Cartesian & non-Cartesian reference frames
- Example of 2D metric tensor

Distance between two points in a **non-Cartesian** frame in 2D



$\{\vec{e}_x, \vec{e}_y\}$ form a Cartesian reference frame $\vec{e}_x \equiv \hat{i}$
 $\vec{e}_y \equiv \hat{j}$

$\{\vec{e}'_x, \vec{e}'_y\}$ do not form a Cartesian frame & do not have unit length

In the Cartesian frame $\{\vec{e}_x, \vec{e}_y\}$

$$Q = (Q_x, Q_y) = (1/2, 0)$$

$$P = (P_x, P_y) = (0, 1/2)$$

$$L = |\vec{Q} - \vec{P}| = \sqrt{(Q_x - P_x)^2 + (Q_y - P_y)^2} = \frac{1}{\sqrt{2}}$$

In non-Cartesian frame $\{\vec{e}'_x, \vec{e}'_y\}$

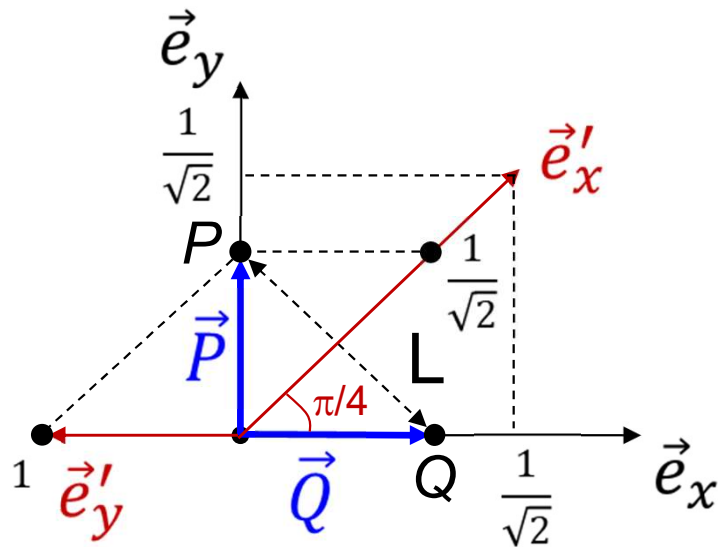
Pythagoras' theorem

$$Q' = (Q'_x, Q'_y) = (0, -1)$$

$$P' = (P'_x, P'_y) = \left(\frac{1}{\sqrt{2}}, 1\right)$$

$$L' = \frac{3}{\sqrt{2}} \neq L \quad \text{incorrect}$$

Distance between two points in a **non-Cartesian** frame in 2D



$\{\vec{e}_x, \vec{e}_y\}$ form a Cartesian reference frame $\vec{e}_x \equiv \hat{i}$
 $\vec{e}_y \equiv \hat{j}$

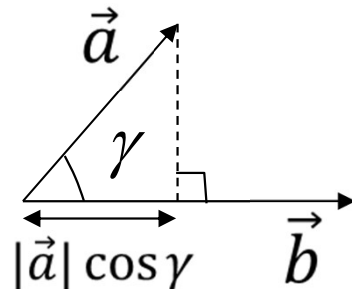
$\{\vec{e}'_x, \vec{e}'_y\}$ do not form a Cartesian frame & do not have unit length

Express the old basis vectors in terms of the new ones...

Visual inspection: $\vec{e}_x = -2\vec{e}'_y$ $\vec{e}_y = \sqrt{2}\vec{e}'_x + 2\vec{e}'_y$

Needs to be generalized...

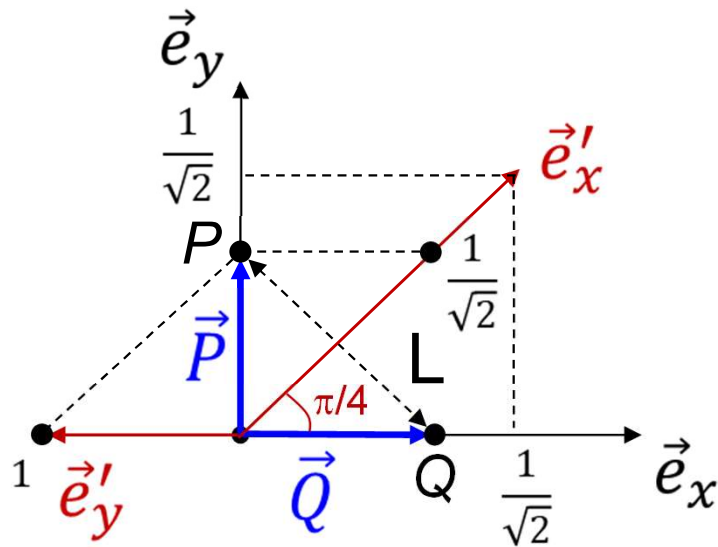
$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \gamma$$



$$\vec{a} \cdot \vec{a} = |\vec{a}| |\vec{a}| \cos 0 \quad |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}$$

$$L = |\vec{Q} - \vec{P}| = \sqrt{(\vec{Q} - \vec{P}) \cdot (\vec{Q} - \vec{P})}$$

Distance between two points in a **non-Cartesian** frame in 2D



$\{\vec{e}_x, \vec{e}_y\}$ form a Cartesian reference frame $\vec{e}_x \equiv \hat{i}$
 $\vec{e}_y \equiv \hat{j}$

$\{\vec{e}'_x, \vec{e}'_y\}$ do not form a Cartesian frame & do not have unit length

$$L = |\vec{Q} - \vec{P}| = \sqrt{(\vec{Q} - \vec{P}) \cdot (\vec{Q} - \vec{P})}$$

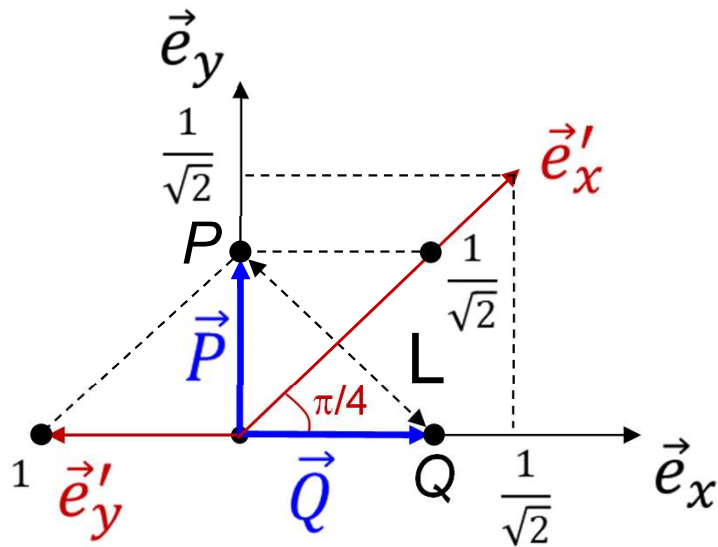
$$\vec{P} = p_x \vec{e}_x + p_y \vec{e}_y \quad \vec{Q} = q_x \vec{e}_x + q_y \vec{e}_y$$

$$L = \sqrt{(q_x - p_x)^2 \vec{e}_x \cdot \vec{e}_x + (q_y - p_y)^2 \vec{e}_y \cdot \vec{e}_y + 2(q_x - p_x)(q_y - p_y) \vec{e}_x \cdot \vec{e}_y}$$

Cartesian case: $\vec{e}_x \cdot \vec{e}_x = 1$; $\vec{e}_y \cdot \vec{e}_y = 1$; $\vec{e}_x \cdot \vec{e}_y = 0$

$$L = \sqrt{(q_x - p_x)^2 + (q_y - p_y)^2} \quad \text{Pythagoras' theorem}$$

Distance between two points in a **non-Cartesian** frame in 2D



$\{\vec{e}_x, \vec{e}_y\}$ form a Cartesian reference frame $\vec{e}_x \equiv \hat{i}$
 $\vec{e}_y \equiv \hat{j}$

$\{\vec{e}'_x, \vec{e}'_y\}$ do not form a Cartesian frame & do not have unit length

$$L = |\vec{Q} - \vec{P}| = \sqrt{(\vec{Q} - \vec{P}) \cdot (\vec{Q} - \vec{P})}$$

$$L = \sqrt{(q_x - p_x)^2 \vec{e}_x \cdot \vec{e}_x + (q_y - p_y)^2 \vec{e}_y \cdot \vec{e}_y + 2(q_x - p_x)(q_y - p_y) \vec{e}_x \cdot \vec{e}_y}$$

Non-Cartesian: $\vec{e}'_x \cdot \vec{e}'_x = 1$; $\vec{e}'_y \cdot \vec{e}'_y = \frac{1}{4}$; $\vec{e}_x \cdot \vec{e}_y = \frac{\cos \frac{3\pi}{4}}{2} = \frac{-1}{2\sqrt{2}}$

$$L' = \sqrt{(q'_x - p'_x)^2 + \frac{1}{4}(q'_y - p'_y)^2 - \frac{1}{\sqrt{2}}(q'_x - p'_x)(q'_y - p'_y)}$$

$$Q' = (Q'_x, Q'_y) = (0, -1)$$

$$P' = (P'_x, P'_y) = (\frac{1}{\sqrt{2}}, 1)$$

$$L' = \frac{1}{\sqrt{2}} = L$$

Correct

Metric tensor in 2D

Dot product between basis vectors

$$L^2 = (q_x - p_x)^2 \vec{e}_x \cdot \vec{e}_x + (q_y - p_y)^2 \vec{e}_y \cdot \vec{e}_y + 2(q_x - p_x)(q_y - p_y) \vec{e}_x \cdot \vec{e}_y$$

$$L^2 = [q_x - p_x \quad q_y - p_y] \begin{bmatrix} \vec{e}_x \cdot \vec{e}_x & \vec{e}_x \cdot \vec{e}_y \\ \vec{e}_y \cdot \vec{e}_x & \vec{e}_y \cdot \vec{e}_y \end{bmatrix} \begin{bmatrix} q_x - p_x \\ q_y - p_y \end{bmatrix} \quad \text{Result is a scalar}$$

Distance between two points, valid in all non-orthonormal reference frames!

e.g., Cartesian: $\vec{e}_x \cdot \vec{e}_x = 1$; $\vec{e}_y \cdot \vec{e}_y = 1$; $\vec{e}_x \cdot \vec{e}_y = 0$

$$L^2 = [q_x - p_x \quad q_y - p_y] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_x - p_x \\ q_y - p_y \end{bmatrix} \quad L = \sqrt{(q_x - p_x)^2 + (q_y - p_y)^2}$$

e.g., Non-Cartesian: $\vec{e}'_x \cdot \vec{e}'_x = 1$; $\vec{e}'_y \cdot \vec{e}'_y = \frac{1}{4}$; $\vec{e}_x \cdot \vec{e}_y = \frac{-1}{2\sqrt{2}}$

$$L'^2 = [q'_x - p'_x \quad q'_y - p'_y] \begin{bmatrix} 1 & \frac{-1}{2\sqrt{2}} \\ \frac{-1}{2\sqrt{2}} & 1 \end{bmatrix} \begin{bmatrix} q'_x - p'_x \\ q'_y - p'_y \end{bmatrix}$$

$$L' = \sqrt{(q'_x - p'_x)^2 + \frac{1}{4}(q'_y - p'_y)^2 - \frac{1}{\sqrt{2}}(q'_x - p'_x)(q'_y - p'_y)}$$